Inferential Statistics

Principles of Information Quality

INFQ 70303

John R. Talburt

Inferential Statistics

- **Inferential statistics -** the set of methods used to make a generalization, estimate, prediction, or decision on a population based on collected samples.

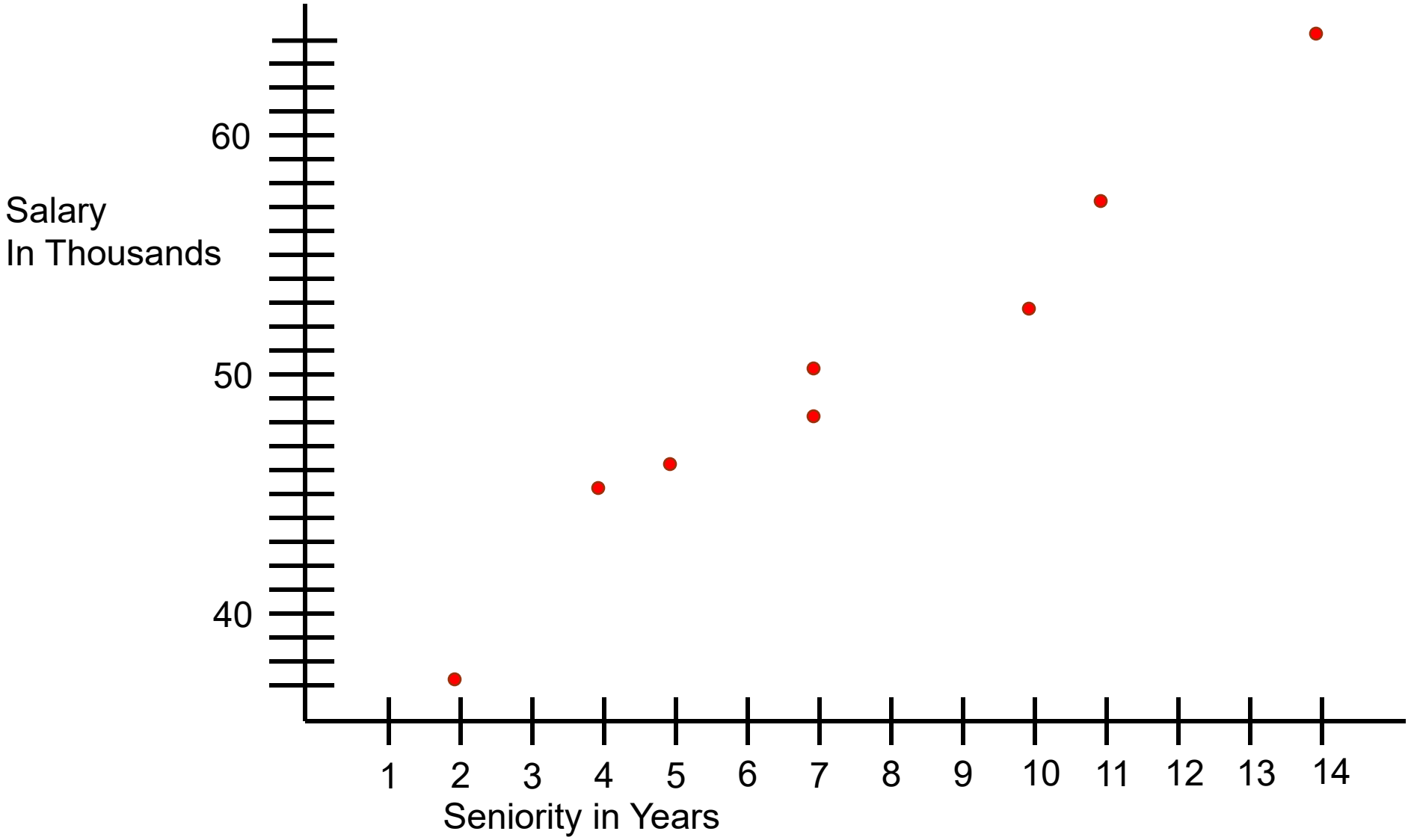
- Statistical inference is the relationship between a population and samples drawn from the population.

Linear Correlation and Regression

- Example

X=Seniority Years	Y=Salary, Thousands
2	37
4	45
5	46
7	48
7	50
10	52
11	56
14	64

Graphical Representation



Linear Correlation Coefficient

- Pearson Product-Moment Coefficient – Quantitative Measure of Linear Correlation
- $C_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{(n-1)s_x s_y}$
- $\bar{x} = Average\ of\ X$
- $\bar{y} = Average\ of\ Y$
- $s_x = Std.Dev.of\ X$
- $s_y = Std.Dev.of\ Y$
- $-1.0 \leq C_{xy} \leq 1.0$
- $0.8 \leq C_{xy} \leq 1.0$ indicate high positive prediction
- $-1.0 \leq C_{xy} \leq -0.8$ indicate high negative prediction

Correlation Coefficient of Example

- Pearson Product-Moment Coefficient
- $C_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{(n-1)s_x s_y}$
- Excel
  "=CORREL(a1:a6, b1:b6)"
- $C_{xy} = 0.9759$
- High Positive Prediction

X=Seniority Years	Y=Salary, Thousands
2	37
4	45
5	46
7	48
7	50
10	52
11	56
14	64

Linear Prediction Formula (Linear Regression)

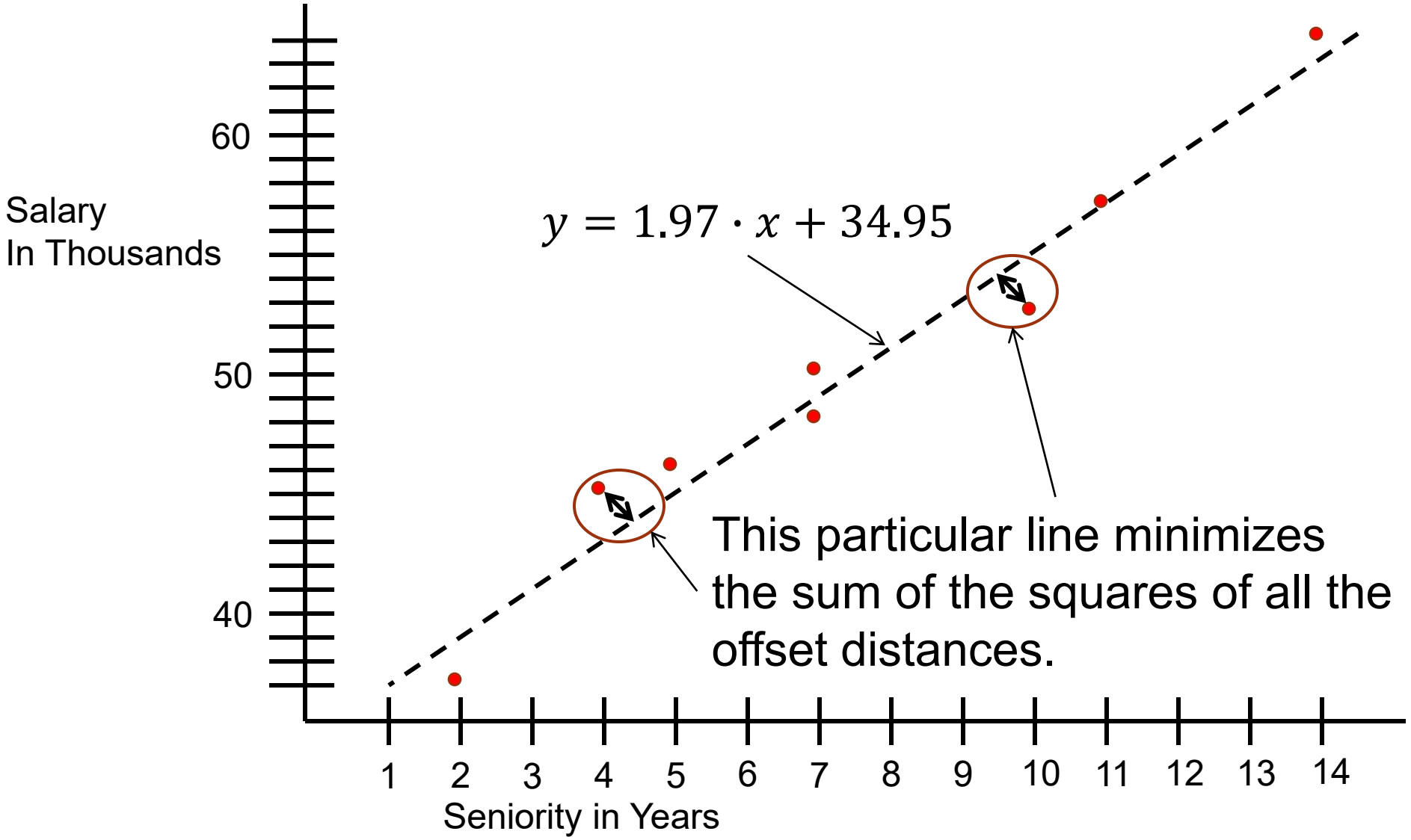
- Model with a Straight Line
- $y = Ax + B$
- $A = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$
- A = 1.97

  Excel

  "=LINEST(Y1:Y6, X1:X6)"
- $B = \bar{y} - A \cdot \bar{x}$
- B = 34.95
- Y = 1.97*x +34.95

X=Seniority Years	Y=Salary, Thousands	Predicted Value of Y
2	37	38.9
4	45	42.8
5	46	44.8
7	48	48.8
7	50	48.8
8	Not Given	50.7
10	52	54.7
11	56	56.7
14	64	62.6
16	Not Given	66.5

Graphical Representation



Sampling Theory and Point Estimates

- **Point estimate** - A single number determined from a sample and used to estimate the population value.

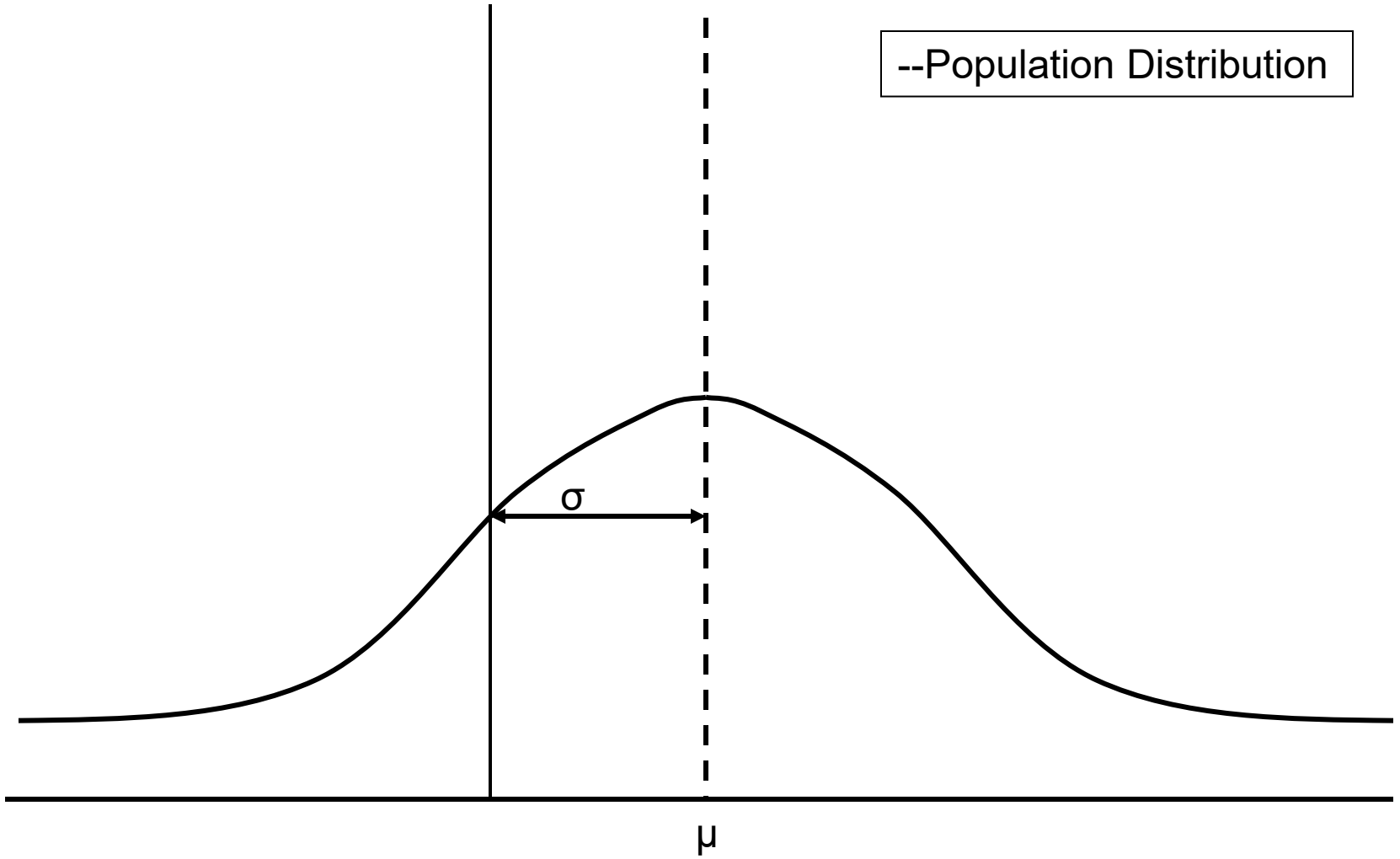
- **Sampling error** - The difference between the sample statistic and the population parameter.

Central Limit Theorem

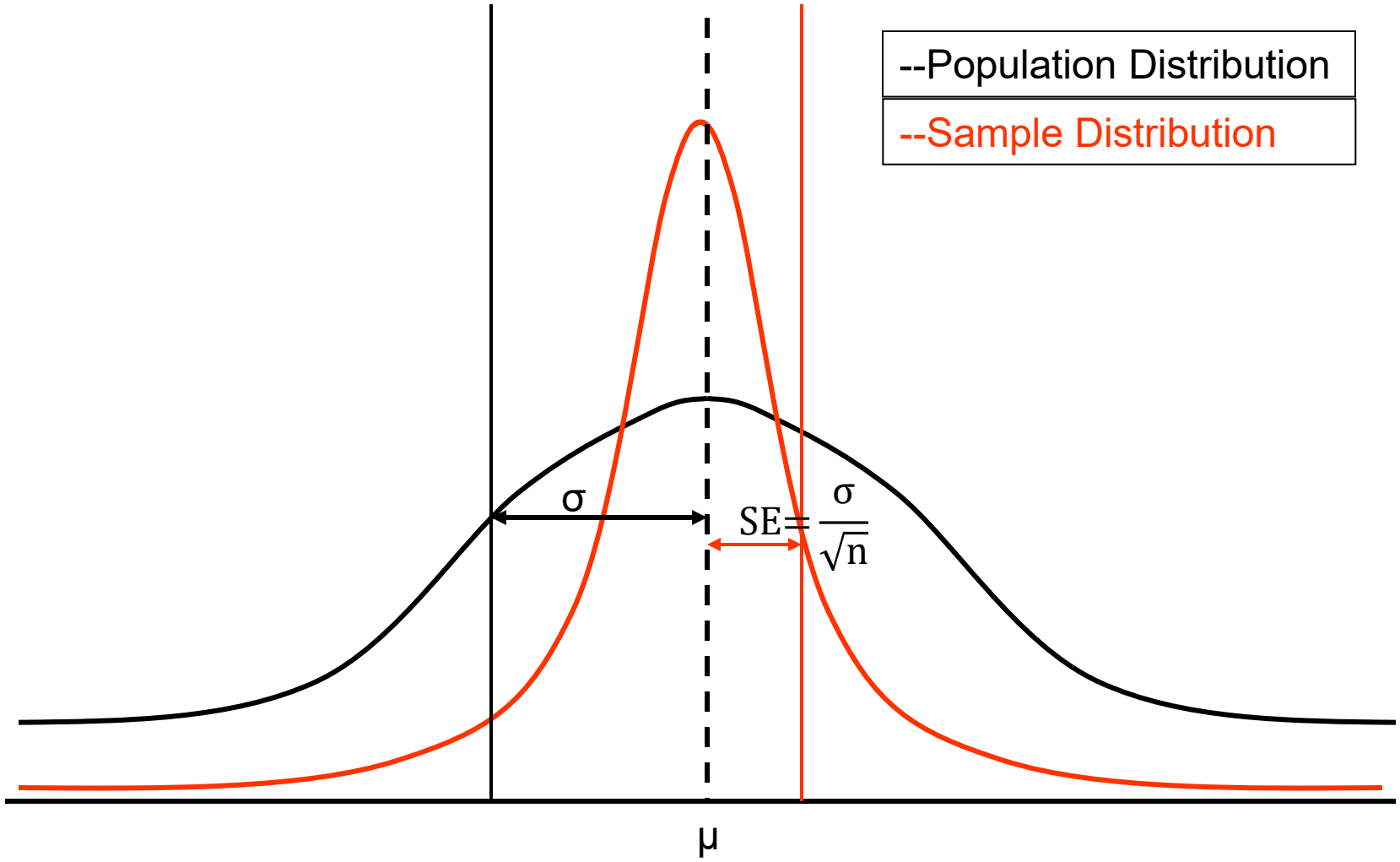
- Central Limit Theorem makes the following claim about the distribution of the sample means:
- In random samples of size **n**, and with increasing values of **n**,
  - The sample means tend to be normally distributed with a mean equal to the population mean and
  - A standard deviation of

$$SE = \frac{\sigma}{\sqrt{n}}$$

Normally Distributed Population



Central Limit Theorem



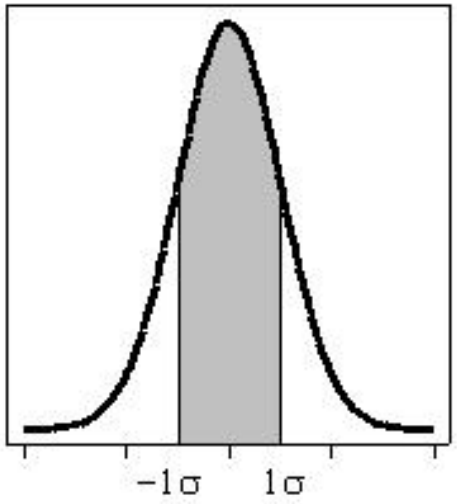
Example

- The manager of a large candle manufacturing plant estimates that on average, each household spends $20/year buying candles. A research firm interviews 300 households at random about their candle purchases. Their data show
  - Mean $16/year purchases
  - Std Dev $10
- Does this sample support or contradict the managers estimate of $20/year?

Analysis

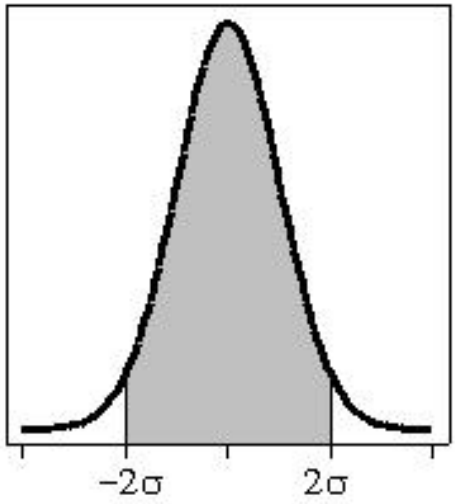
- If the candle purchases are normally distributed about a mean value of $20 , then we expect samples of 300 to be normally distributed about a mean value of $20 with
  - SE = σ/Sqrt(300) $\approx$ 10/Sqrt(300) = 0.577
  - Z = (16 – 20)/0.462 = -6.93
- Probability that sample mean differs from population mean by almost 7 standard deviations is almost zero.
  - Better Guess is $\mu = 16 \pm 2(0.577)$
  - Approximately 95% chance μ is between $14.84 and $17.15

Z-Scores in Normal Distributions



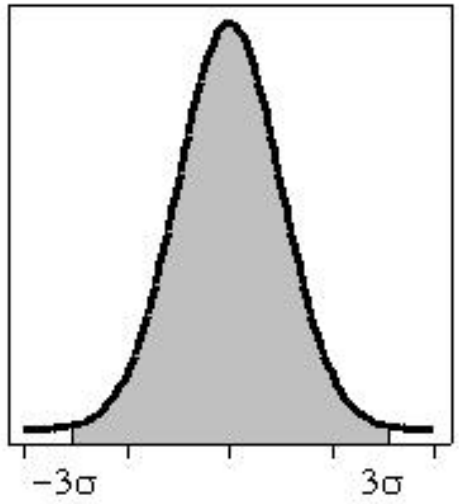
Z (in std. deviation units)

(a) shaded area= 68.27%



Z (in std. deviation units)

(a) shaded area= 95.45%



Z (in std. deviation units)

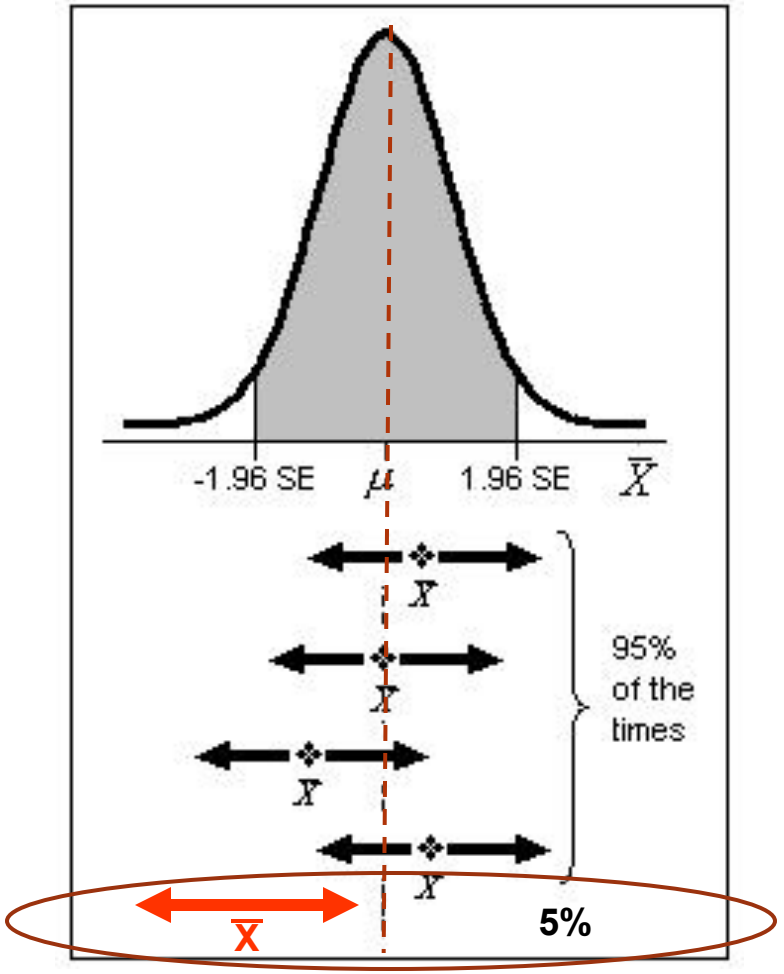
(a) shaded area= 99.73%

Confidence Levels and Z-scores

Confidence Level	68.27%	90.00%	95.00%	95.45%	99.00%	99.73%
Interval Coefficient (IC) (Z-score)	1	1.645	1.96	2	2.58	3

Some characteristic Z-scores and their corresponding confidence levels.

Confidence Intervals



Confidence Intervals

Confidence Interval Estimates

- The **interval estimate** is an estimate of a population parameter given by two numbers (lower and upper bound) between which a population parameter probably lies.

- The **confidence interval** for the population parameter is the interval that has a high probability of containing the population parameter.

Confidence Intervals and Tolerance Limits

- An interval estimate is analogous to tolerance limits - provides upper and lower bounds (limits) for a target measurement.

- For example, a 95% confidence interval implies that about 95% of the time, and interval of this size will enclose the parameter being estimated.

Confidence Interval

- The general formula for the confidence interval:

  Interval = Point Estimate ± (Interval Coefficient) x SE

$$\left[\left(\bar{X} - IC \cdot \frac{\sigma}{\sqrt{n}}\right), \left(\bar{X} + IC \cdot \frac{\sigma}{\sqrt{n}}\right)\right]$$

- The ***interval coefficient (IC)*** is the number of standard errors on either side of the population parameter necessary to include a percentage of the point estimate equal to the confidence level.

Estimated Std Dev

- If the population standard deviation is not known, the sample standard deviation can be used provided the sample size is large (≥ 30)

- In this case

$$SE = \mathbf{s}/\mathrm{Sqrt}(n)\ ,$$

where **s** is the standard deviation of the sample

Z-Score and Interval Coefficient

When the

1. Sample size is large ($n \geq 30$)
2. The population standard deviation is known or is estimated by the sample standard deviation
3. The sampling distribution of the statistic is approximately normal,

Then the interval coefficient is a Z-score
(standard deviation units)

The choice of the Z-score depends on confidence level.

Example

- Time required to complete a quality check is normally distributed with std dev of 2 min. If a sample of 100 checks has a mean of 10.5 min, then what will be a 95% confidence interval?
- $\bar{X} = 10.5, \quad \sigma = 2, \quad$ 95% interval coefficient = 1.96
- SE = 2/Sqrt(100) = 0.2
- SE * 1.96 = 0.2*1.96 = 0.392
- $10.5-0.392 \leq \mu \leq 10.5+0.392$
- $10.108 \leq \mu \leq 10.892$
- Excel Function "CONFIDENCE(0.05, 2, 100)"

Hypothesis Testing

Hypothesis Testing

- **Hypothesis testing –** A claim about the parameter of a population under consideration, and setting up a test procedure in which a sample statistic is tested to verify the claim.

- **Null hypothesis (H_0)** - Hypothesis or claim being tested.

- **Alternative hypothesis (H_a)** – Contains any value not contained in the null hypothesis.

Example: Hypotheses

- Auditor wants to test the claim that the average A/R balance is $500. Std Dev is known to be $50. Significance level of test should be 5% (α, *probability of rejecting true null hypothesis*)

- H_0 is that $\mu = \$500$

- H_a is that $\mu \neq \$500$

Example: Test

- Pull a sample of 49 accounts ($n = 49$)
- Sample mean is \$475 ($\bar{X} = 475$)
- Construct a 95% confidence interval around the **hypothesis mean** ($\mu = 500$)
- SE = σ/Sqrt(n) = 50/7 = 7.14
- Z score for 95% = 1.96
- Interval is $\mu \pm (1.96 \cdot 7.14)$ or 500 ± 13.99
- $W_{C1} = 486.01$ and $W_{C2} = 513.99$
- Reject H_0 because $475 \leq [486.01, 513.99]$

Sampling Distribution of Differences of Proportions and Means

Population Parameter	Point Estimate	Standard Error
Mean (μ)	$\bar{X}$	$\frac{\sigma}{\sqrt{n}}$
Proportion (π)	p	$\sqrt{\pi \cdot (1-\pi)/n}$
Difference between means of two populations (μ_1-μ_2)	$\bar{X}_1 - \bar{X}_2$	$\sqrt{\frac{(\sigma_1)^2}{n_1} + \frac{(\sigma_2)^2}{n_2}}$
Difference between proportions of two populations (π_1 - π_2)	$p_1 - p_2$	$\sqrt{\frac{\pi_1 \cdot (1-\pi_1)}{n_1} + \frac{\pi_2 \cdot (1-\pi_2)}{n_2}}$

Proportion Example

- Proportion of Defects in a population is 0.1 (10%)
- If random samples of size 50 are taken, then
- SE = SQRT((0.1 *(1-0.1))/50) = .04
- interval coefficient = 2 gives 95.45%
- 2 x SE = 0.08
- Expect that more than 95% of the time that the proportion of defects found in a random sample of 50 items will be between 0.02 (0.1-0.08) and 0.18 (0.1+0.08)

Questions or Comments